H2 Math Summary

Example Questions

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1 Systems of Linear Equations

1.1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	Discounts given for each package			Total price after the discount
	Blend A	Blend B	Blend C	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		Comments
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

1.2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		**Comments**
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

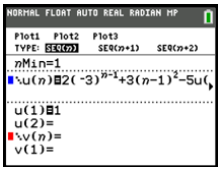
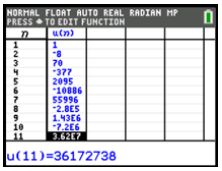
1.3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		**Comments**
[2]	$n = 1$ $x_2 = a(-3)^1 + b(1)^2 + cx_1$ $-8 = -3a + b + c \quad ---(1)$ $n = 2$ $x_3 = a(-3)^2 + b(2)^2 + cx_2$ $70 = 9a + 4b - 8c \quad ---(2)$ $n = 3$ $x_4 = a(-3)^3 + b(3)^2 + cx_3$ $-377 = -27a + 9b + 70c \quad ---(3)$ Using GC, $a = 2,\ b = 3,\ c = -5$ $x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$ Using GC, $x_{10} = -7\,210\,868$ $x_{11} = 36\,172\,738$   	

1.4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

Solution		**Comments**
(i) **[2]**	$1-2\lambda = 2\mu \quad ---(1)$ $2+\lambda = 5-2\mu \quad ---(2)$ $2\lambda = -8+3\mu \quad ---(3)$ Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$ Sub into (3): $\text{LHS} = 2(-2) = -4$ $\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$ $\text{LHS} \neq \text{RHS},$ no solution $\therefore l_1$ and l_2 do not intersect $\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$ $\therefore l_1$ and l_2 are not parallel $\therefore l_1$ and l_2 are skew (shown)	

1.5 A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

Age group	Discounted ticket price
$\leqslant 12$ years	\$6
13 to 49 years	\$9
$\geqslant 50$ years (senior citizen)	\$4

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

[TJC 2022 Prelim P2 Q1]

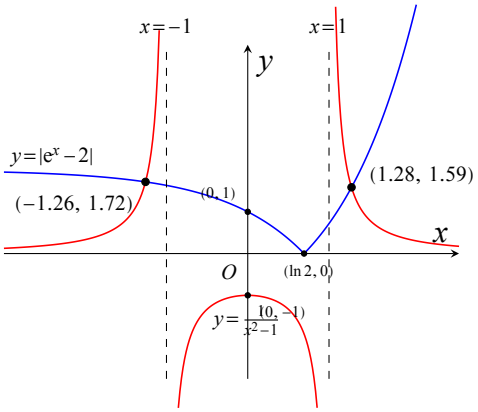
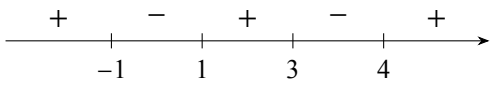
Solution		**Comments**
[4]	Let x, y, z be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively. $\left.\begin{aligned} x+y+z=100 \quad &---(1) \\ 6x+9y+4z=700 \quad &---(2) \end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$ $z \geqslant 35$ $\because x \geqslant 0$ $\frac{200}{3}-\frac{5}{3}z \geqslant 0$ $z \leqslant 40$ $\therefore 35 \leqslant z \leqslant 40$ x, y, z are non-negative integers. Using GC table: $z=37 \Rightarrow x=5,\ y=58$ $z=40 \Rightarrow x=0,\ y=60$ $\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$	

2 Inequalities

2.1 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

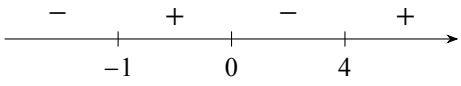
Solution		Comments
(a) **[3]**	 Using GC, intersections at $x=-1.2581$, $x=1.2770$ $\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$	
(b) **[3]**	$\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$ $\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$ $\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$ $\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$  $\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$	

2.2 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

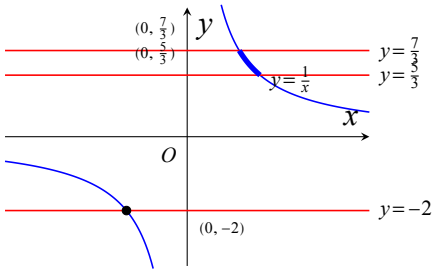
[ACJC 2026 Prelim P1 Q1]

Solution		**Comments**
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\frac{x(5x+9)}{x^2+2x+5} > x$ $5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1$ or $0<x<4 \quad ---(2)$ $(1)\cap(2)$: $\therefore x<-1$ or $0<x\leqslant 3$	

2.3 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

[SAJC 2025 Prelim P2 Q1]

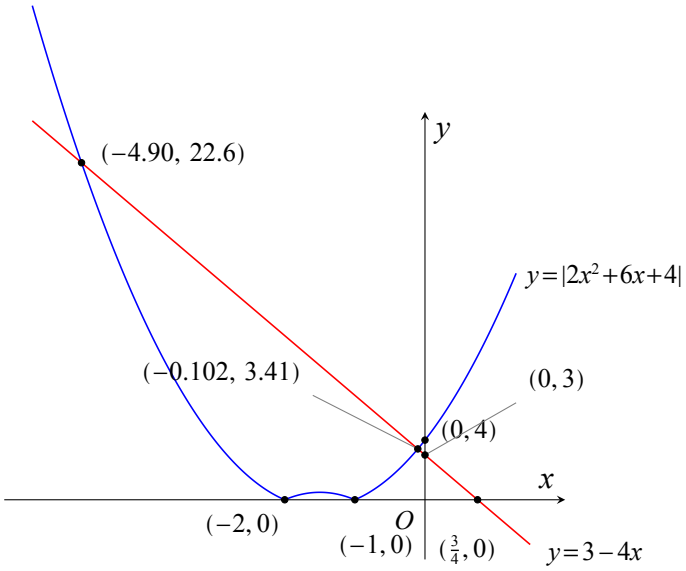
Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2$: $0 \leqslant 0$ (true) $x \neq -2$: $\|x+2\|>0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$ $3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	

2.4 **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

(ii) Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

[SAJC 2019 Prelim P1 Q2]

Solution	Comments
(i) **[4]**	



$$3-4x=2x^2+6x+4$$

$$2x^2+10x+1=0$$

$$x=\frac{-5\pm\sqrt{23}}{2}$$

$$\therefore x<\frac{-5-\sqrt{23}}{2} \text{ or } x>\frac{-5+\sqrt{23}}{2}$$

(ii) **[3]**

$$\text{Area}=\int_{-3}^{-1}(3-4x-|2x^2+6x+4|)\,dx$$

$$=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$$

$$=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$$

$$=\left[-\tfrac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\tfrac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$$

$$=\left(\tfrac{16}{3}-20+2\right)-(18-45+3)+\left(-\tfrac{2}{3}+1-7\right)-\left(-\tfrac{16}{3}+4-14\right)$$

$$=20$$

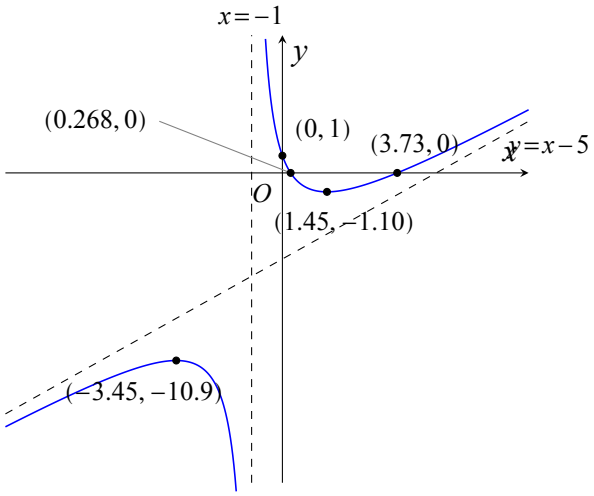
3 Graphing Techniques

3.1 A curve C has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

(a) Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

(b) Sketch C, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

[VJC 2025 Promo Q9(a), (b)]

Solution		Comments
(a) **[3]**	$y=\dfrac{x^2-4x+1}{x+1}$ $yx+y=x^2-4x+1$ $x^2+(-y-4)x+(1-y)=0$ No real solution for x: $(-y-4)^2-4(1-y)<0$ $y^2+8y+16-4+4y<0$ $y^2+12y+12<0$ $(y+6)^2-24<0$ $\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$ $-6-2\sqrt{6}<y<-6+2\sqrt{6}$ $\therefore y$ cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$	
(b) **[4]**	$y=x-5+\dfrac{6}{x+1}$ Asymptotes: $x=-1,\ y=x-5$ $x=0$: $y=1$ $y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$ Turning points: $(1.45,-1.10),\ (-3.45,-10.9)$  	

3.2 A curve C has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where a and b are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

(a) If C has no stationary points, use differentiation to find the relationship between a and b. [3]

(b) It is now given that $b=a$.

(i) Find the x-coordinates of the turning points. [2]

(ii) Hence sketch C, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

(iii) Verify that the point of intersection of the two asymptotes of C lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of k in terms of a, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

[SAJC 2025 Promo Q7]

Solution		Comments
(a) **[3]**	$\dfrac{dy}{dx}=a-\dfrac{2b-a}{(x+3)^2}$ $\dfrac{dy}{dx}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$ $(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$ No stationary points $\Rightarrow \dfrac{2b-a}{a}<0$ Since $a>0,\quad 2b-a<0$ $\therefore a>2b$	
(b)(i) **[2]**	$b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$ $x+3=1$ or -1 $x=-2$ or -4	

(b)(ii)
[3]

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

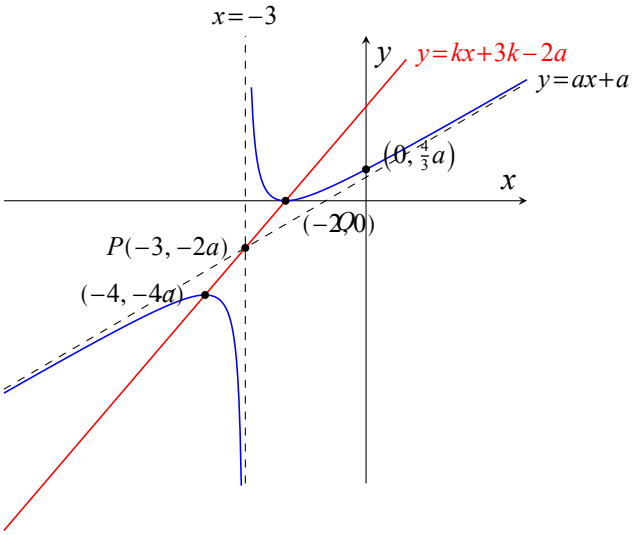
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$



(b)(iii)
[2]

$x = -3$ and $y = ax + a : \quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3 : \quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through P meets $C \Rightarrow k > a$

3.3 A curve C has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and k is a non-zero real constant.

It is given that the function f is strictly increasing.

(a) Find the range of values of k. [3]

(b) Sketch the curve C, giving the coordinates of any points where C crosses the x- and y-axes and the equations of any asymptotes. [3]

(c) By adding a suitable curve to the graph of the curve C in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

[DHS 2025 Promo Q8]

Solution		Comments
(a) **[3]**	$\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$	
	$\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$	
	f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$	
	$3x^2+12x+12-k>0$ $\quad\because (x+2)^2>0$ for $x>-2$	
	$3(x+2)^2-k>0$	
	Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$	
	$k\neq 0 \Rightarrow k<0$	
(b) **[3]**	$y=3x+\dfrac{k}{x+2}$	
	Asymptotes: $y=3x,\ x=-2$	
	$x=0:\ y=\dfrac{k}{2}$	
	$y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$	
	$x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$	

(c)
[3]

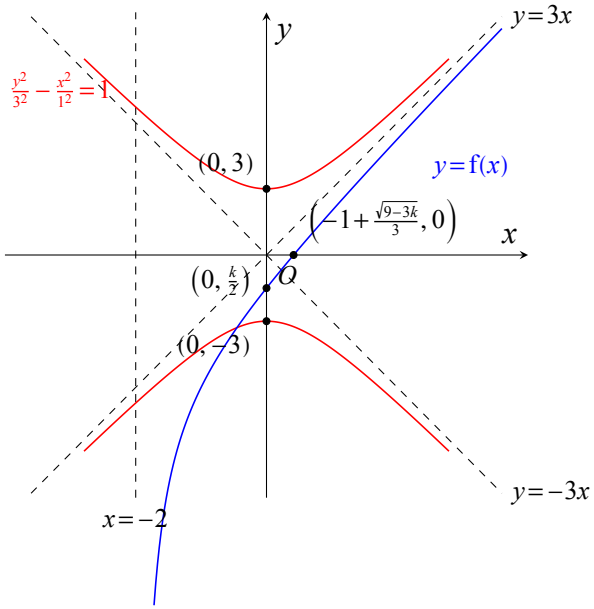
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$



$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

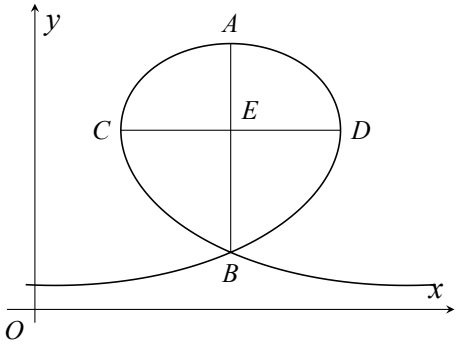
$\therefore$ $(*)$ has one real root

3.4 Engineers are designing a roller coaster. The diagram shows the side view of a section of track with a loop. This section is modelled by the curve with parametric equations

$$x = 28(\theta \cos\theta + 1), \qquad y = 20(2 - \theta \sin\theta),$$

for $-\frac{2}{3}\pi \leqslant \theta \leqslant \frac{2}{3}\pi$, where x and y are measured in metres. The x-axis models the horizontal ground.

The maximum point on the curve is at A, and the line AB is vertical. Points C and D are where the tangent to the curve is parallel to the y-axis. Point E is the intersection of the lines AB and CD.



(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of θ. [3]

(b) **(i)** Find the values of θ at points C and D, giving your answers correct to 2 significant figures. [3]

(ii) Hence find the greatest width, CD, of the loop. [2]

The x-coordinate of point A is 28.

(c) **(i)** Find the value of θ at point A. [1]

(ii) Find the 2 possible values of θ at point B. [2]

(iii) Hence find the distance AB. [1]

Engineers know that, for safety reasons, the distance AE should be within 10% of the distance $\frac{1}{2}CD$.

(d) Determine whether the design of this section of track satisfies this condition. [2]

[A-Level 2025 P1 Q11]

Solution		Comments
(a) **[3]**	$x=28(\theta\cos\theta+1) \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}\theta}=28(-\theta\sin\theta+\cos\theta)$ $y=20(2-\theta\sin\theta) \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}\theta}=20(-\theta\cos\theta-\sin\theta)$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{20(-\theta\cos\theta-\sin\theta)}{28(-\theta\sin\theta+\cos\theta)}=\dfrac{5(\theta\cos\theta+\sin\theta)}{7(\theta\sin\theta-\cos\theta)}$	
(b)(i) **[3]**	At C and D: $\theta\sin\theta-\cos\theta=0$ Using GC, $\theta_C=-0.86$, $\theta_D=0.86$ (2 s.f.)	
(b)(ii) **[2]**	$CD=x_D-x_C$ $=28(\theta_D\cos\theta_D+1)-28(\theta_C\cos\theta_C+1)$ $=31.4$ (3 s.f.)	
(c)(i) **[1]**	A is a maximum point: $\dfrac{\mathrm{d}y}{\mathrm{d}x}=0$ $5(\theta\cos\theta+\sin\theta)=0$ $\therefore \theta=0$	
(c)(ii) **[2]**	$x=28$: $28(\theta\cos\theta+1)=28$ $\theta\cos\theta=0$ $\theta=0$ or $\cos\theta=0 \Rightarrow \theta=0, \pm\dfrac{\pi}{2}$ $\theta=0$ is point $A \Rightarrow \theta=\pm\dfrac{\pi}{2}$	
(c)(iii) **[1]**	$AB=y_A-y_B$ $=20(2-0)-20\left(2-\dfrac{\pi}{2}\right)$ $=31.4$ (3 s.f.)	
(d) **[2]**	$AE=y_A-y_D=40-20(2-\theta_D\sin\theta_D)=13.043$ $\left(0.9\times\frac{1}{2}CD,\ 1.1\times\frac{1}{2}CD\right)=(14.139,\ 17.282)$ $AE\notin(14.139,\ 17.282) \Rightarrow$ the design does not satisfy the condition	